

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1 Which factor most fundamentally limits the performance of rational agents?

- a) Infinite computational resources
- b) Deterministic environments
- c) Perfect sensory inputs
- d) Reactive architectures

Rationality of an agent depends on:

- a) Human-like behavior
- b) Past environment states only
- c) Performance measure, percept sequence, agent's knowledge, and actions
- d) Internal emotions and subjective reasoning

3 A heuristic function $h(n)$ is admissible if:

- a) $h(n) = 0$ for all n
- b) $h(n)$ overestimates the true cost
- c) $h(n)$ never overestimates the actual cost to reach the goal
- d) $h(n)$ is always equal to $g(n)$

Alpha-Beta pruning is primarily used in:

- a) Constraint Satisfaction Problems
- b) Local Search optimization
- c) Game-playing adversarial search
- d) Uninformed search strategies

What is a key advantage of using predicate logic over propositional logic?

- a) Faster resolution
- b) Ability to express relationships between objects
- c) Simpler truth table representation
- d) Requires less computation

6 In knowledge representation, procedural knowledge is typically harder to:

- a) Represent
- b) Store
- c) Explain
- d) Execute

7 **Bayesian networks model uncertainty through:**

- a) Conditional probability distributions
- b) Fuzzy membership functions
- c) Logic predicates
- d) State-space representation

8 **Semantic analysis in NLP focuses on:**

- a) Sentence boundary detection
- b) Parsing sentence structure
- c) Extracting meaning from text
- d) Managing discourse relations

Which form of machine learning typically involves learning from a set of pre-classified examples?

- a) Reinforcement Learning
- b) Unsupervised Learning
- c) Inductive Learning
- d) Genetic Algorithms

10 **Expert systems differ from standard AI systems mainly because they:**

- a) Focus solely on probabilistic reasoning
- b) Use case-based reasoning exclusively
- c) Capture human domain knowledge in rule-based structures
- d) Do not require inference engines

A neural network's weights are best described as:

- a) Random variables representing uncertainty
- b) Adjustable parameters that encode learned knowledge
- c) Logical operators
- d) Static control knowledge

Which local search strategy is most susceptible to getting stuck in local optima?

- a) Hill Climbing
- b) Simulated Annealing
- c) Genetic Algorithms
- d) Tabu Search

Q2. Solve the following.

- A) Discuss the historical evaluation of Artificial Intelligence.
- B) State the relationship between agent and environment.

Q3. Solve the following.

- A) Give the outline of uniform cost search algorithm.
- B) What is alpha-beta pruning? Explain the function of alpha-beta pruning.

Q4. Solve Any Two of the following.

- A) What is planning? What is need of planning.
- B) Give outline of simple forward Chaining algorithm with suitable example.
- C) Explain the production based knowledge representation technique.

Q5. Solve Any Two of the following.

- A) Explain Bayesian Network.
- B) Compare and contrast the following:
Forward Vs. Backward Reasoning

- C) What is fuzzy set? Explain degree of membership. How does are represent a fuzzy set in a computer. 6

Q6. Solve Any Two of the following.

- A) Comparison between Syntactic Processing & Semantic Processing. 6
B) What is Expert system? Explain its Architecture, Features & applications. 6
C) What is Natural Language Processing? Explain various types of NLP techniques. 6

*** End ***